Chapter 14 exercises • Protection

OS14-E01 • Domains and least privilege

Phase A needs only read(S); B only write(T); C parses untrusted input.

1. What excess does a static union domain expose? Define dynamic restrictions.
2. Does user mode alone isolate these phases?

OS14-E02 • Exact matrix updates

Initially A[D1, F]={r*, owner}, A[D2, F]={w}, A[D3, F]=∅; D1 also controls D2.

1. Limited-copy r to D3, then transfer r* to D2. Show F's column after each.
2. Which two rights can remove D2's w? Can control add x to D2?

OS14-E03 • Reachable domains versus simultaneous rights

D1→D2→D3; only D3 can print P. Current domain is D1.

1. Can it print immediately? Can it print later legally?
2. Delete D2→D3 with no other path. Prove printing impossible for every subsequent switch-only trace.

OS14-E04 • ACLs, capabilities, and existing opens

F ACL=[U1:r, U2:rw], G ACL=[U2:r, U3:w], allow-only/default-deny.

1. Convert to user capability rows.
2. U1 already has a read-only open handle. After deleting its ACL entry, can it write or read?

OS14-E05 • Generation numbers and ABA

Cap=(slot5, g3, r), table5=(F, g3, live). Revoke F, then reuse slot5 for G.

1. Explain slot-only failure and rejection with generations.
2. With a two-bit counter and an everlasting old token, after how many reuse steps does g3 recur? Give a remedy.

OS14-E06 • Active RBAC roles

U is assigned Backup and Grade. They cannot be active in the same session. Backup grants readSubmissions; Grade grants writeGrades.

Chapter 14 exercises • Protection (continued)

1. Evaluate writeGrades under active={Backup}. What must switching check?
2. Does another session with Grade violate the stated rule?

OS14-E07 • Historical Java inspection

Use simplified SE 8 without explicit/limited context. Stack N→L→A has p at N/L, not A; inherited context holds p.

1. Evaluate ordinary inspection and L doPrivileged.
2. Remove L's p. Does privileged marking help? What if all ordinary frames hold p but inherited context does not?
3. What fails when L accepts arbitrary addr and connects in a privileged block?

OS14-E08 • Delegation and information flow

D1 holds F:r and G:w, but not F:r*.

1. Can it grant D2 read authority? Can it copy F's data into G?
2. Why does attenuation induction not directly apply to a system with owner rights?

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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https://github.com/xinhi6716/computer-science-self-study